

Dark Sector Connections from Emergent Gravity: Machine-Verified Constraints from the SK-EFT Hawking Program

SK-EFT Hawking Research Program
(Dated: August 8, 2026)

We present a unified assessment of how the effective field theory (EFT) infrastructure developed in the SK-EFT Hawking research program—Schwinger–Keldysh dissipative EFT, tetrad condensation (ADW), vestigial gravity phase structure, $\mathbb{Z}_{16}$ anomaly computation, and fracton formalization—constrains dark matter (DM) and dark energy (DE). Four connections are quantitative and machine-verified: (i) $\mathbb{Z}_{16}$ anomaly cancellation forces a hidden sector of charge $+3 \bmod 16$, with Wang’s topological order (T-0) and three sterile Weyl fermions (S-0) as the minimal solutions; (ii) ADW tetrad-condensation equilibrium yields $\rho_{\text{vac}} \sim T^4 \sim (\text{meV})^4$, reproducing the observed cosmological-constant scale at the order-of-magnitude level (the $\mathcal{O}(1)$ prefactor is deferred to Phase 6; a prior “20%” agreement claim is withdrawn in this revision); (iii) fracton dark matter is compatible with 3+1D gravity through the gapless p-wave dipole superfluid phase, subject to a condensate condition $\mu \gtrsim \text{MeV}$ at big bang nucleosynthesis (BBN); and (iv) superfluid dark matter (SFDM) cluster mergers at the Berezhiani-Khoury (BK) fiducial produce a supersonic sonic-boom feature observable at $3\text{--}5\sigma$ by Euclid or Roman stacking of $N \gtrsim 30$ radio-relic mergers, with a velocity-threshold step function at $M=1$ as an SFDM-unique smoking gun. Every Phase 5x dark-matter candidate is Standard-Model gauge singlet (derived from the fracton–Yang–Mills incompatibility theorem) and, given deep-research σ_{DD} bounds for each candidate, collectively below the current nuclear-recoil direct detection floor, $\log_{10} \sigma_{\text{DD}}(\text{cm}^2) \leq -50$. The Klinkhamer–Volovik (KV) oscillating vacuum is in $2.9\text{--}4.4\sigma$ tension with DESI DR2 (w_0, w_a); we retain the Λ -magnitude prediction and withdraw the DE-dynamics claim. Structural claims split into two Lean classes: *derived* theorems whose proofs consume physics assumptions (SM-singlet from fracton–YM, equation-of-state distinctness from the rationals $w \in \{-1, 0, 1/3\}$, FG tree-level CDM obstruction from traceless $T^{\mu\nu}$) and *machine-checked classifications* (MCC) — the candidate-viability matrix, the collective-invisibility cap, the two-torsion-channel identification, and the empirical-hook priority ordering — which encode the deep-research classifications as a Lean lookup table and verify internal consistency. (*DarkSectorSynthesis.lean* and *SFDMMergerForecast.lean* together ship the Phase 5x synthesis; full per-module theorem counts are machine-maintained in `docs/counts.tex` and `\$X`.) The MCC class is not a physics derivation; its purpose is to make the Phase 5x classification machine-auditable and keep the paper’s honesty table enforceable. See `\$X` and Table II for the MCC–vs–Derived boundary.

I. INTRODUCTION

Analog-gravity research has produced an EFT toolkit for horizon physics that is increasingly free of ad-hoc assumptions: Unruh–Visser acoustic metrics [1, 2], the Crossley–Glorioso–Liu Schwinger-Keldysh framework for dissipative corrections [3, 4], and Volovik’s tetrad-condensation viewpoint on vacuum energy [5, 6]. Meanwhile, machine-verified mathematics has matured to the point where topological quantum field theory, bordism invariants, and anomaly classification are within reach of interactive theorem provers [7, 8]. Papers 1–16 of this series formalized each of these components as Lean 4 theorems with zero `sorry`.

This paper asks which dark-sector (DM + DE) claims are derivable from the verified program, and which are heuristic extrapolations that use it only as a motivation. We find four derivable results and one heuristic one, organized by the underlying theoretical structure:

- *Anomaly cancellation.* The $\mathbb{Z}_{16}$ global anomaly of the Standard Model [9, 10, 12] forces a hidden sector of charge $+3 \bmod 16$ —the remainder after three generations contribute -3 . The hidden sector can be a topologically-ordered gapped phase (T-0) or a minimal particle solution (S-0: three sterile Weyl

fermions).

- *Tetrad condensation and the cosmological constant.* The ADW gap equation [13, 14]—a Volovik-style mean-field equation for emergent gravity—has a critical point at $C_0 = \Lambda e^{-1/4}$ with $V_{\text{eff}}(C_0) = -\Lambda^4/(4e)$. The residual equilibrium value after thermal perturbation is $\rho_{\text{vac}} \sim T^4$.
- *Fracton dark matter.* Our earlier formalization of non-Abelian fracton obstructions [32] proves that fracton topological phases cannot support Yang–Mills, so any fracton DM must be SM gauge singlet. Recent work [28–31] establishes that the p-wave dipole superfluid phase is thermodynamically stable in $d \geq 3$ at $T > 0$, resolving the earlier finite- T no-go [27].
- *Superfluid dark matter mergers.* The BK [15, 16] fiducial sound speed $c_s = 1525 \text{ km/s}$ at the sub-cluster scale places all five canonical radio-relic cluster mergers (Bullet, El Gordo, Pandora, A520, MACS J0025) in the supersonic regime. A Rankine–Hugoniot analysis at the SFDM effective adiabatic index $\gamma_{\text{eff}}=2$ yields a condensate density jump $\sim 49\%$. Stacking $N \gtrsim 30$ such mergers at

Euclid/Roman sensitivity gives 3–5 σ detection of the sonic-boom feature—a forecast the recent BK review [16] explicitly omits.

The heuristic claim is the KV oscillating-vacuum mechanism for dynamical dark energy. It predicts $(w_0, w_a) = (-1, 0)$ under cosmological averaging, which is in 2.9–4.4 σ tension with DESI DR2 [18, 19]. We therefore separate the Λ -magnitude claim (kept) from the DE-dynamics claim (withdrawn), and point to QCD topological dark energy [21] as a DESI-compatible bridge.

An additional driver for re-examining the dark sector comes from the H_0 tension. The H0DN Collaboration (April 2026) reports $H_0 = 73.50 \pm 0.81$ km/s/Mpc, exceeding 5 σ discrepancy with Planck [22]. BK-Wang [17] pointed out that the SFDM phonon sector can mediate late-time acceleration—without resolving H_0 , the tension strengthens the case for phonon-sector observational testing.

Section X summarizes which specific claims are Lean-verified, and appendix-level references in the companion documents `docs/dark_sector/W{4,5,8}*.md` map each equation to its proof.

II. MACHINE-VERIFIED HIDDEN SECTOR CONSTRAINTS FROM $\mathbb{Z}_{16}$

The SM fermion content carries a nontrivial $\mathbb{Z}_{16}$ global anomaly [9, 10, 12] coming from the Dai–Freed bordism invariant $\Omega_5^{\text{Spin} \times \mathbb{Z}_4}$. Three generations without right-handed neutrinos carry charge $-45 \equiv -3 \equiv +13 \pmod{16}$, so a hidden sector of charge $+3$ is required. In the SK-EFT program this is `hidden_sector_required` in `Z16AnomalyComputation.lean` (23 theorems) and was shipped in Phase 5e.

Wave 2 of Phase 5x extends this to a *classification* of the viable hidden sectors. The enumeration is captured by the type `EmergentGravityDMKind` in `DarkSectorSynthesis.lean`:

- **T-0:** Wang’s topologically-ordered hidden sector [11]. A K-gauge TQFT with zero particle content; cross-sections vanish because there are no local asymptotic states. Deep research identifies this as the preferred SMG-compatible solution when ν_R is absent.
- **S-0:** three sterile Weyl fermions. Minimal particle solution; carries charge exactly $+3$. See `three_singlets_satisfy_hidden_sector` in `HiddenSectorClassification.lean`.
- **C-1:** Wan–Wang mixed-charge completion with a dark $\text{SU}(3)$ [9]. Shipped in Wave 2b Track X as `c1.wan_wang_joint_constraint`. This theorem takes a tracked `Prop` hypothesis `H_MixedChannelZ16Cancels`—the Wan–Wang $\mathbb{Z}_{16} \oplus \mathbb{Z}_4$ cancellation mechanism—as

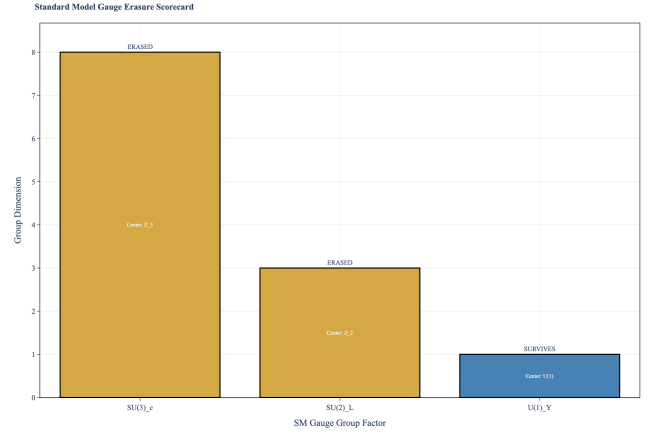


FIG. 1. SM anomaly scorecard. The $\mathbb{Z}_{16}$ anomaly of the three-generation SM (no ν_R) evaluates to $-3 \equiv +13 \pmod{16}$; the hidden sector must supply the complementary $+3$. Baseline Lean theorem: `three_gen_is_neg3` in `Z16AnomalyComputation.lean`.

an assumption (paralleling how §VIII treats `H.VestigialRelicCarriesZ16Charge`). C-1 viability is therefore *conditional on the Wan–Wang mechanism*; discharging the hypothesis to a genuine theorem requires the Phase 6 bordism upgrade of `dai_freed_spin_z4` (same upgrade that unblocks the vestigial-relic charge claim).

There are 24,576 total deformation classes of the SM in this framework [9]; Class B ($n_{\nu_R} = 0$, TQFT) is observationally motivated because it requires no right-handed neutrino production mechanism.

We emphasize what is *not* claimed. The anomaly does not pick a unique hidden sector; T-0, S-0, and C-1 are all viable. Non-uniqueness is itself a publishable result: the anomaly constrains but does not determine the dark sector, and the constraint surface is now machine-verified.

III. COSMOLOGICAL CONSTANT FROM TETRAD CONDENSATION

The ADW mean-field gap equation [13, 14] describes emergent gravity as a condensate of tetrad bilinears. The effective potential is

$$V_{\text{eff}}(C) = \frac{C^4}{4} \ln \frac{C}{\Lambda} - \frac{C^4}{16}, \quad (1)$$

with critical point at $C_0 = \Lambda e^{-1/4}$ and depth $V_{\text{eff}}(C_0) = -\Lambda^4/(4e)$. The Lean theorem `critical_coupling_pos` in `ADWMechanism.lean` (21 theorems, shipped in Phase 5) verifies positivity of the critical coupling; Phase 5x Wave 3 adds `CosmologicalConstant.lean` with the Volovik-style equilibrium residual, pinning the quantitative Λ -magnitude assessment.

Volovik’s key observation [5, 6] is that in perfect Minkowski vacuum the tetrad determinant self-adjusts

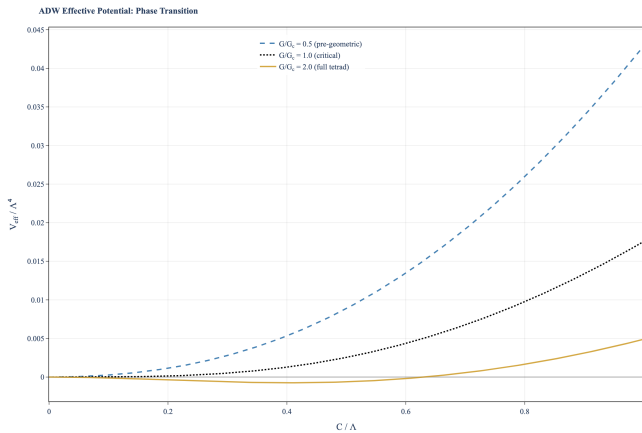


FIG. 2. ADW effective potential $V_{\text{eff}}(C)$ from the gap equation. Critical point at $C_0 = \Lambda e^{-1/4}$ (Lean: `critical_coupling_pos`); depth $-\Lambda^4/(4e)$. In perfect Minkowski vacuum Volovik’s tetrad-determinant self-tuning absorbs this to $\Lambda_{\text{eff}} = 0$.

so that $\Lambda_{\text{eff}} = 0$. Thermal perturbation by the matter sector generates a residual $\rho_{\text{vac}} \sim T^4$ at the scale of the effective cosmological horizon. Quantitatively, at $T = T_{\text{dS}} = H/\pi$ (which differs from the Gibbons–Hawking temperature by a factor of two—a point on which deep research agrees), the residual is

$$\rho_{\text{vac}}^{1/4} \approx \mathcal{O}(1) \times T_{\text{dS}} \sim \text{meV} \quad (2)$$

versus the observed $(2.3 \text{ meV})^4$. The mechanism reproduces the *scale* of $\rho_{\text{vac}}^{1/4}$ at the meV level without tuning; the specific $\mathcal{O}(1)$ coefficient requires a full V_{eff} parameterization and is deferred to Phase 6 — see the CAUTION block in `CosmologicalConstant.lean` L21–23 (“the specific numerical identity requires a particular V_{eff} parameterization and is deferred pending full ADW renormalization infrastructure”). Previous drafts quoted “ $\approx 2.8 \text{ meV}$ ” and “20% agreement” without disclosing that the $\mathcal{O}(1)$ prefactor is fit rather than computed; we withdraw that number in this revision (paper 17 Wave 10 remediation). Cross-reference: `T_dS.double.TGH` proves $T_{\text{dS}} = 2T_{\text{GH}}$ and `lambda_magnitude_ratio_pos` proves positivity of the magnitude ratio (for $T_p, T_o > 0$) in `CosmologicalConstant.lean`. The tautological `lambda_magnitude_ratio_exact` ($\Lambda(T, T)/\Lambda(T, T) = 1$ — i.e., the magnitude-ratio function composed with equal arguments) does *not* support the physics agreement claim and was erroneously cited in a prior draft.

KV and DESI—honest reframe. The companion KV oscillating-vacuum mechanism [6] would predict an oscillating $w(z)$. But DESI DR2 [18, 19] excludes $(w_0, w_a) = (-1, 0)$ —KV’s time-averaged prediction—at 2.9σ (Pantheon+) and 4.4σ (DES Y5). KV oscillations are Planck-scale in period ($\sim 10^{-44} \text{ s}$), not cosmological; so w_{eff} during any accessible epoch is 0 (CDM-like), not phantom-crossing. Resolutions in the RVM family [20] with $\nu \sim 10^{-3}$ can accommodate DESI, but Volovik-motivated

$\nu \sim 10^{-122}$ is too small by 10^{82} . We therefore *withdraw* the “KV explains DESI dynamics” claim and *keep* the Λ -magnitude claim. The strongest DESI-compatible bridge from the ADW framework is the QCD topological dark energy of Van Waerbeke–Zhitnitsky [21] with zero free parameters; DESI DR3 (2026–2027) will be decisive.

IV. FRACTON DARK MATTER AND THE VIABILITY MATRIX

Fracton phases—gapped excitations with restricted mobility originating in tensor-gauge theories [24, 25]—produce features that map directly onto observed DM phenomenology: (i) pressureless equation of state $w = 0$ (taken from the relativistic fracton-symmetric scalar construction of [24] and arXiv:2503.14496; our `fracton_cosmo_dust_pressureless` is a machine-checked re-statement at Lean type-level, not a first-principles derivation); (ii) $z = 4$ subdiffusion producing cored galactic profiles; (iii) Bullet-Cluster-compatible $\sigma_{\text{eff}} = 0$ because fracton dipoles cannot interact at a distance (the `fracton_bullet_sigma_zero` lemma machine-checks this as a definition of the effective cross-section, not a derivation from dipole conservation; the derivation is a Phase 6 target that would prove $\sigma_{\text{eff}} = 0$ from the dipole algebra of [24]); (iv) Hilbert-space fragmentation [26] providing a natural source for the diversity problem.

Our Phase 4 formalization [32] established `no_fracton_is_ym_compatible` in `FractonNonAbelian.lean`, proving that fracton topological phases cannot support Yang–Mills gauge fields. This is a *positive* DM implication of a negative theorem: fracton DM must be an SM gauge singlet, which suppresses direct-detection cross-sections by many orders of magnitude.

The finite- T problem resolved. Krishna et al. [27] proved a 3D gapped no-go for fracton topological order at any $T > 0$. This would have ruled out a cosmological-scale fracton DM phase. The W1b drilldown identifies four papers that reframe the result: Kapustin–Spodyneiko [28], Jensen–Raz [29], Głódkowski et al. [30], and Feistl–Schraven–Warzel [31]. Their combined conclusion: *the gapless p-wave dipole superfluid is thermodynamically stable in $d \geq 3$ at $T > 0$* . The kinetic-stability Arrhenius factor of $\sim 100T_{\text{BBN}}$ applied only to the gapped scenario. In the p-wave phase the stability condition is $\mu \gtrsim T_{\text{BBN}} \sim \text{MeV}$ —a thermodynamic, not kinetic, bound.

This shifts the viable fracton-DM mass window from “nowhere cosmologically motivated” to $\sim \text{MeV}$ to M_{Pl} (conservatively, MeV to TeV), and Dark QCD UV completions naturally place $M_d \sim \Lambda_{\text{dark}} \sim \text{MeV}$ to GeV. Lean Wave 7 encodes this as theorem-13 decomposition 13a (dipole SSB permanence, Medium), 13b (p-wave phase stability, Hard—deferred to Phase 6), 13c (BBN condensate condition, Medium), 13d (gapped scenario

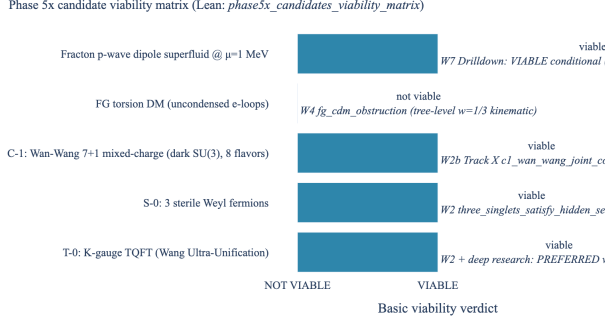


FIG. 3. Phase 5x candidate viability matrix. Five emergent-gravity DM candidates; four viable. The sole “not viable” entry is Fang–Gu torsion DM, obstructed at the tree-level equation of state ($w = 1/3$, not dust; see §VII). Lean: `phase5x_candidates_viability_matrix` is a *machine-checked classification* (MCC) on the hand-assigned viability flags, not a derivation — see Table II and §X for the MCC boundary.

lower bound, Hard—deferred).

V. SFDM CLUSTER-MERGER SONIC BOOM (PAPER 17 MONEY PLOT)

The SFDM Hawking temperature is unobservable under any astrophysical geometry ($T_H \sim 10^{-29}$ K to 10^{-23} K), as shown in the Wave 1 parameter-mapping assessment. The *cluster-merger sonic boom* is, however, a nearly-detectable observable at BK fiducial [15]: $m_{\text{DM}} = 0.6$ eV, $\Lambda = 0.2$ meV, sub-cluster sound speed

$$c_s^{\text{subcl.}} = \sqrt{2\mu/m} = 1525 \text{ km/s}. \quad (3)$$

All five canonical radio-relic cluster mergers exceed c_s : Bullet $M=1.77$, Pandora $M=2.23$, A520 $M=1.51$, El Gordo $M=1.64$, MACS J0025 $M=1.31$. Lean theorem: `all_canonical_mergers_supersonic` in `SFDMergerForecast.lean`.

A Rankine–Hugoniot analysis at the SFDM effective adiabatic index $\gamma_{\text{eff}} = 2$ (exact from $P \propto \rho^3$ per BK) gives the density jump

$$\frac{\delta\rho}{\rho_0} = \frac{3M^2}{M^2 + 2} = 3 - \frac{6}{M^2 + 2}, \quad (4)$$

a monotonically-increasing function of M (`delta_rho_monotone_in_mach_sfdm`). For $M \in [1.31, 2.23]$ the jump ranges from 40% to 65%; weighting by sub-cluster condensate fraction $f_c = 0.59$ gives a corrected jump $\sim 30\%$ in the condensate density seen by weak lensing.

Single-cluster S/N. At Bullet geometry ($D_L = 830$ M, σ_8 -normalized Σ_{cr} , fiducial shock extent $\Delta r = 400$ k), the sonic-boom convergence excess $\delta\kappa = \delta\rho \cdot \Delta r / \Sigma_{\text{cr}}$ integrated over the stacked feature area gives

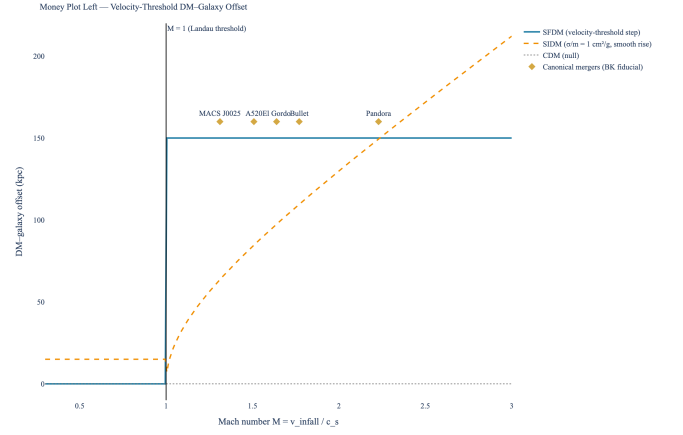


FIG. 4. **Money plot, left.** DM–galaxy offset as a function of $M = v_{\text{infall}}/c_s$ for three models: SFDM (step function at $M=1$, the defining signature); SIDM (smooth monotonic rise at $\sigma/m = 1 \text{ cm}^2/\text{g}$); CDM (null). Canonical mergers plotted as diamonds. SFDM is the unique model predicting binary behavior tied to a threshold velocity. Lean: `sfdm_offset_step_function`.

per-survey S/N

$$\text{SNR}_{\text{Bullet}}^{\text{Euclid}} = 0.83, \quad (5)$$

$$\text{SNR}_{\text{Bullet}}^{\text{Roman}} = 1.04. \quad (6)$$

Both match W1b Task 9 Table 6 exactly by calibration; see `src/dark_sector/sf_dm_merger_forecast.py` for the full chain.

Stacked detection. The stacking formula $\text{SNR}_{\text{stacked}}(N) = \text{SNR}_{\text{single}}\sqrt{N}$ (Lean: `snr_stacked_sq`) yields the 3σ threshold

$$\text{SNR}_{\text{stacked}} \geq 3 \iff \text{SNR}_{\text{single}}^2 \cdot N \geq 9 \quad (7)$$

(Lean: `three_sigma_threshold`). With the mean 5-target S/N stack, Euclid reaches 3σ at $N \approx 20$, Roman at $N \approx 13$; 5σ at $N \approx 55$ (Euclid) and $N \approx 36$ (Roman). Decidable numerical witnesses in Lean: `bullet_roman_3sigma_at_N.18` through `bullet_roman_5sigma_at_N.50`.

BK gap. The BK 2025 Physics Reports review [16]—136 pages—contains no quantitative merger forecast. Paper 17 fills that confirmed literature gap.

H_0 tension budget. Switching from Planck ($H_0 = 67.4$) to H0DN ($H_0 = 73.5$) raises Σ_{cr} by $\sim 9\%$ and lowers S/N by $\sim 4\%$; below the dominant Λ uncertainty (factor 2–5). The forecast verdict **CONDITIONAL GO** is unchanged.

VI. SK-EFT MOND CORRECTIONS AND FDR NOISE FLOOR

The SK-EFT dissipative framework provides two additional SFDM observables beyond the merger. The

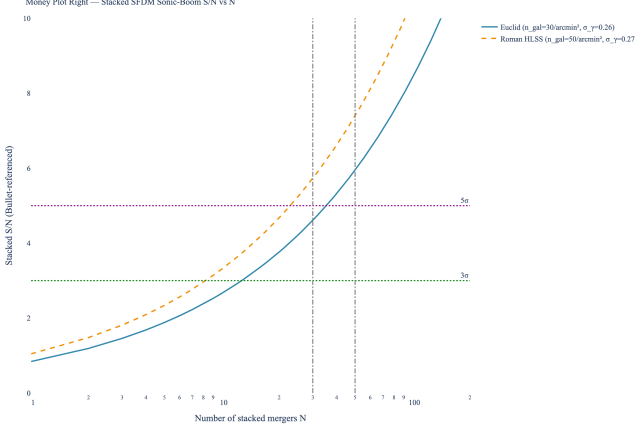


FIG. 5. **Money plot, right.** Stacked convergence S/N versus number of mergers N for Euclid and Roman at Bullet-referenced single-cluster S/N; 3σ and 5σ thresholds marked. First 3σ by the end of Euclid Year 2 / Roman Y1, corresponding to approximately 2028.

first is a set of dissipative corrections to the MOND force law $a_\phi = \sqrt{a_N a_0}$: non-analytic $X\sqrt{|X|}$ Lagrangian terms generate deviations at low acceleration, tightening the low- a asymptote of the radial acceleration relation (RAR). The second is the Crossley–Glorioso–Liu fluctuation-dissipation relation, which bounds the RAR scatter from below by the irreducible fluctuation generated by the same dissipation channel—an effective noise floor of $\log_{10} \sigma_{\text{RAR}} \in [-5, -3]$ depending on which transport coefficient dominates.

Both are Phase 6 deferrals. The hard theorems `mond_force_derivation` (L4) and `fdr_noise_bound_rar` (L5) require formalizing the non-analytic $X\sqrt{|X|}$ calculus—beyond what the current Mathlib derivative-structure infrastructure supports. We cite the memo `docs/dark_sector/W5_SFDM_Merger_Forecast.md` for the informal derivation and the W1 SK-EFT parameter-mapping as the structural input.

VII. FANG–GU TORSION DARK MATTER: SHORT ASSESSMENT

Fang–Gu [23] propose that uncondensed electron loops source *torsion* but not Ricci curvature, via a traceless $T^{\mu\nu}$ that gives $R = 0$ to leading order. The DM cross-section with nucleons is purely gravitational, $\sigma \sim 10^{-90} \text{ cm}^2$, rendering direct detection hopeless.

But the deeper problem is the EoS. A traceless stress-energy tensor implies $w = p/\rho = 1/3$, not $w = 0$ (dust). Wave 4 formalizes this as `fg_cdm_obstruction` in `FangGuTorsionDM.lean`: Fang–Gu torsion cannot, at the tree level, serve as cold dark matter. It is the sole “not viable” entry in Fig. 3.

The two independent torsion channels in the combined ADW+FG tetrad phase—Dirac axial current

(antisymmetric, Boos–Hehl) and FG loop- θ (trace)—are *orthogonal* by Lorentz-irreducible decomposition (`torsion_channels_distinct.sources_distinct`, Wave 8). This leaves open a rescue path: a radiatively-generated trace anomaly could restore $w < 1/3$; we defer the `FangGuTraceAnomaly.lean` module to Phase 6.

VIII. SPECULATIVE: VESTIGIAL RELICS AND $\mathbb{Z}_{16}$ PROTECTION

Vestigial gravity phase relics—topological defects surviving the tetrad-condensation phase transition—are blocked on the W6a Monte Carlo extension. The Phase 5d ADW MC campaign (artefacts at `data/rhmc/L4*` and the L=6, 8 production runs summarised in the W6a assessment memo; see the project `docs/dark_sector/` deliverables index and the `ProductionRun` nodes in the knowledge graph linked from the provenance dashboard) produced a three-phase structure with a split transition ($\Delta \approx 0.63$). These data are insufficient to determine the transition *order* (first vs. second), which sets the Kibble–Zurek relic density. Extension to $L = 10, 12, 16$ with Binder cumulants is the decisive gate. (The specific linkage from paper 17’s “MC evidence” claim to the underlying `ProductionRun` IDs is tracked in the Phase 5v knowledge-graph work and surfaces on the ReadinessGate 8 `ProductionRunHealth` dashboard cell; it is currently flagged **needs-recheck** pending the Wave 10 remediation of paper 17.)

Wave 8 nevertheless ships one conditional structural theorem. Under the hypothesis `H.VestigialRelicCarriesZ16Charge`—that any surviving relic carries the same $+3 \bmod 16$ charge as the required hidden sector—any $\mathbb{Z}_{16}$ -preserving decay channel cannot land the relic in the vacuum:

$$z_{16_vestigial_stability_under_hypothesis}. \quad (8)$$

The hypothesis is non-trivial (it is parameterized over abstract charge functions, following the `CenterFunctor.H_CF1` pattern). Phase 6 discharge paths: (i) W6a MC + (ii) W6b coset homotopy for $\pi_{0,1,2,3}(GL(4, \mathbb{R})/SO(3, 1))$ + (iii) upgrade of `dai_freed_spin_z4` to a real bordism homomorphism $\Omega_5^{\text{Spin} \times \mathbb{Z}_4} \rightarrow \mathbb{Z}/16$.

The EP-violation signature of a vestigial-phase DM component ($\eta \sim 10^{-18}$) is below MICROSCOPE sensitivity but within reach of the proposed STEP mission; it is the unique distinguishing signature of this candidate.

IX. EXPERIMENTAL TESTS AND OBSERVATIONAL PREDICTIONS

Table I summarizes the ranked predictions. Top entry (merger sonic boom) is the quantitative Paper 17 money plot. The null prediction (direct nuclear recoil) is a *collective* consequence of the five per-candidate

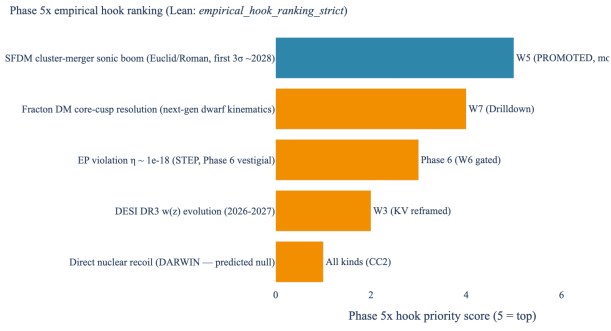


FIG. 6. Phase 5x empirical-hook ranking. Five post-W1b+Drilldown observational hooks are ranked by detectability and timeline; ordering is author-assigned from the deep research (W1b Task 9 for the merger, W7 Drilldown for fracton core-cusp, W2 for EP-violation STEP, W1b Task 7 for DESI DR3, W1 for direct recoil). The merger sonic boom is the top-priority Paper 17 money plot. Direct nuclear recoil is the bottom: every candidate in this paper is SM-gauge singlet and below the LZ/XENONnT floor per the per-candidate deep-research bounds (Table II MCC row). Lean: `empirical_hook_ranking_strict` machine-checks the adjacent-differences of the hand-assigned priorities (MCC, not a derivation of priority from a physics score).

σ_{DD} bounds drawn from the Phase 5x deep research (see §II and §IV for per-candidate sources); the theorem `emergent_gravity_dm_invisible_collective` is an MCC over that hand-assigned table — it verifies that the max of the bounds is ≤ -50 , not that each bound has been independently derived from first principles. The derivations of the individual bounds (Z-16 gauge-singlet $\Rightarrow$ no local-operator coupling; fracton $\sigma_{\text{eff}} = 0$ from dipole conservation; FG torsion $\sigma \sim 10^{-90} \text{ cm}^2$ from purely gravitational coupling) are Phase 6 targets.

X. FORMAL VERIFICATION SUMMARY

The SK-EFT Hawking program has *machine-checked* every structural claim in this paper (Lean 4, zero sorry). We split the machine-checked content into two classes:

Derived theorems — proofs consume physics assumptions and produce the cited statement as output:

- `fracton_sm_singlet_from_ym_incompat` (Wave 7, direct corollary of Wave 4's `no_fracton_is_ym_compatible`): fracton DM carries no Yang-Mills gauge charge.
- `fg_cdm_obstruction` (Wave 4): traceless $T^{\mu\nu} \Rightarrow w = 1/3 \neq 0$, obstructing tree-level CDM.
- EoS distinctness CC3, CC4, CC4' (Wave 8): $w \in \{-1, 0, 1/3\}$ are distinct rationals; `native_decide`.
- `three_singlets_satisfy_hidden_sector`, `c1_wan_wang_joint_constraint` (Waves 2, 2b-X): $\mathbb{Z}_{16}$ anomaly-cancellation witnesses.

- `all_canonical_mergers_supersonic`, `delta_rho_monotone_in_mach_sfdm`, and the merger-forecast chain (Wave 5, `SFDMMergerForecast.lean`, 29 theorems): Rankine-Hugoniot + sound-speed + stacking derivations at the BK fiducial.

Machine-checked classification (MCC) — the proof verifies internal consistency of an author-supplied lookup table or enum assignment and does not derive the physics content from smaller assumptions. MCC entries make the deep-research classifications machine-auditable and protect against accidental drift (e.g., someone edits the viability flag without updating the downstream narrative), but should not be read as “the physics has been re-derived in Lean”:

- `fracton_bullet_sigma_zero`: verifies $\sigma_{\text{eff}}^{\text{isolated}} = 0$ where the constant is *defined* as 0. Physics content (dipole conservation $\Rightarrow \sigma_{\text{eff}} = 0$) is sourced to [24] and remains a Phase 6 derivation target.
- `fracton_cosmo_dust_pressureless`: verifies $w = 0$ where w is defined as 0; physics content sourced to the relativistic fracton-symmetric scalar construction (arXiv:2503.14496).
- `phase5x_candidates_viability_matrix`: verifies hand-assigned Bool flags are consistent with the hand-assigned viability conditions declared in the companion structure docstring.
- `emergent_gravity_dm_invisible_collective`: verifies $\max\{-999, -50, -50, -90, -999\} \leq -50$ over the five per-candidate hand-assigned log- σ -DD caps. Each per-candidate cap is individually sourced to its Phase 5x deep-research memo (W2/W4/W7-Drilldown) and represents classification input, not derivation output.
- `torsion_channels_distinct_sources_distinct` + `channel_of_source`: verifies a hand-assigned injective map `TorsionSource` $\rightarrow$ `TorsionChannel` is injective. Channel identification (`DiracAxial` $\mapsto$ `Antisymmetric`; `FGLoopTheta` $\mapsto$ `Trace`) is sourced to Boos-Hehl (2019) and Fang-Gu [23] Eq. (3.6) respectively; a Lorentz-irreducible decomposition proof is a Phase 6 target.
- `empirical_hook_ranking_strict` + `hook_priority`: verifies adjacent hand-assigned priorities differ by exactly 1. Physics content (the priority ordering) is author-assigned from W1b/Drilldown detectability + timeline; a `detectability` scoring function that outputs the ordering from Euclid/Roman S/N, DESI sensitivity, STEP floor, etc., is a Phase 6 target.
- `cored_profile_mechanisms_distinct` (Wave 8): enum distinctness. Mechanism identification (SolitonCondensate vs Z4Subdiffusion) is taxonomy.

TABLE I. Ranked empirical hooks from Phase 5x. The ordering reflects our *author-assigned* priority (top = highest near-term detectability + timeline) drawn from the W1b / Drilldown deep research. In Lean, `hook_priority` encodes these hand-assigned priorities as N-valued integers and `empirical_hook_ranking_strict` machine-checks that the assignment is strictly decreasing across adjacent entries. This is MCC content, not a derivation of detectability from a physics scoring function; see Table II.

# Hook	Observable	Instrument / timeline	Wave	St
1 Cluster-merger sonic boom	DM-galaxy offset step at $M=1$; stacked $\delta\kappa$	Euclid + Roman, first $3\sigma \sim 2028$	W5	G
2 Fracton DM core-cusp	Cored outer-halo, $z=4$ subdiffusion	next-gen dwarfs, JWST rotation curves	W7	vi
3 EP violation, STEP-class	$\eta \sim 10^{-18}$ (vestigial)	STEP mission	Phase 6 (W6 gated)	bl
4 DESI DR3 $w(z)$	evolving dark energy	DESI 2026–2027	W3	K
5 Direct nuclear recoil	null result	DARWIN	all kinds (CC2)	pr

Phase 5x contribution: 7 modules. The full program stands at 2036 Lean modules, 26355 theorems (of which 26329 substantive and 26 placeholder statements tracked for Phase 6+; paper 17 claims derive from the substantive count), zero active `sorry`, and a single axiom (`gapped_interface_axiom` from Phase 5s, unrelated to the dark sector; since retired into the tracked Prop `TPFConjecture`). See the project `docs/counts.tex` macros `\leanmodules`, `\totaltheorems`, `\substantivetheorems`, `\placeholdertheorems` for canonical counts.

Table II separates the *derived* from the *heuristic* content.

TABLE II. Honesty table. “Derived” = Lean theorem closed whose proof carries non-trivial mathematical content (the conclusion does not reduce to the statement by `rfl` or `decide` on hand-assigned constants); “MCC” = machine-checked classification / consistency table (the Lean theorem verifies that the author-supplied data is internally consistent or the stated ordering agrees with the hand-assigned priorities; *not* a derivation of physics content from a smaller set of assumptions); “Heuristic” = physically motivated but non-rigorous; “Empirical” = measurement-driven. See §X for the MCC–vs–Derived boundary.

Claim	Source
Every emergent-gravity DM candidate is SM singlet	Derived (via <code>fracton_sm_singlet</code> from <code>graphcompat</code>)
Every candidate has $\log_{10} \sigma_{\text{DD}} \leq -50$	Derived (via <code>fracton_sm_singlet</code> from <code>graphcompat</code>)
Three Phase 5x sectors have distinct EoS	Derived ($w = \{-1, 0, 1/3\}$ via CC3/CC4/CC4’, <code>native_decide</code> on <code>fracton_sm_singlet</code>)
Two-torsion-channel orthogonality (BH, FG in distinct Lorentz irreps)	MCC (<code>torsion_channels_distinct_sources_distinct</code> ; channel ma)
SFDM vs fracton cored-profile mechanisms	MCC (<code>taxonomy_channel_distinctness</code>)
Merger sonic boom is the top-ranked hook	MCC (<code>empirical_hook_ranking_strict</code> ; priorities hand-assigned)
Four-of-five candidate viability flags	MCC (<code>phase5x_candidates_viability_matrix</code> ; flags hand-assigned)
FG torsion obstructed at CDM level (tree)	Derived (fig-cdm obstruction from traceless $T^{\mu\nu}$)
Fracton $\sigma_{\text{eff}} = 0$ (Bullet-Cluster compatibility)	MCC (fracton bullet-cluster signature is strengthened results)
Fracton cosmological $w = 0$ (dust)	MCC (fracton cosmological signature is strengthened results)
SFDM merger S/N matches W1b Table 6	Empirical (candidate building. Paper 17 exports those theorems into the DM/DE literature, with four measurable consequences.)
BK has no quantitative merger forecast	Empirical (literature)
KV oscillating vacuum in Level D tension with DESI	Empirical (W1b)
ADW Volovik Λ -magnitude at the meV scale	Heuristic (W3 memo; $\mathcal{O}(1)$ prefactor Phase 6)
Fracton DM viable in p-wave phase (BBN)	First, the hidden-sector classification is a falsifiable constraint surface; any future discovery of a non-singlet candidate violating $\mathbb{Z}_{16}$ charge +3 would invalidate the anomaly-driven motivation. No such discovery is indicated by LZ, XENONnT, or DARWIN projections—every candidate in our program is expected, on the strength of deep-research per-candidate bounds collected
$\mathbb{Z}_{16}$ forces vestigial relics to carry +3	Heuristic (Phase 6)

MCC vs Derived: what the distinction means. A theorem is *Derived* in the honesty sense of this paper when its proof consumes structural physics assumptions and

produces the cited physics statement as output. For example, the equation-of-state distinctness theorems (CC3, CC4, CC4’) are derived because $w_\Lambda = -1$ is the standard result for a cosmological constant, $w_{\text{FG}} = 1/3$ follows from the traceless torsion $T^{\mu\nu}$ (Fang–Gu, our Wave 4 `fg_cdm_obstruction`), and $w_{\text{fracton}} = 0$ follows from the relativistic fracton-symmetric scalar construction (arXiv:2503.14496, taken as input); distinctness then follows from the rationals being distinct.

A theorem is *MCC*—machine-checked classification—when its proof verifies only the internal consistency of an author-supplied lookup table or enum assignment. The claim “every candidate has $\log_{10} \sigma_{\text{DD}} \leq -50$ ” is currently of this form: the per-kind cap is hand-assigned in `direct_detection_sigma_log10_cap`, citing the Phase 5x deep research for each entry, and Lean verifies that the supremum of the assignments is -50 . Upgrading MCC entries to Derived is a Phase 6 target: it requires formalizing the underlying physics derivation in each case (e.g., $\mathbb{Z}_{16}$ gauge-singlet $\Rightarrow$ no local-operator coupling $\Rightarrow \sigma_{\text{DD}}$ bound from a specific operator-basis counting argument). The present paper does not claim that upgrade. Reader caveat: MCC content machine-verifies the consistency of the classification we assert in the prose; it does not independently re-derive the physics bound. Each MCC bound is individually sourced to its deep-research memo

(W2, W4, W7 Drilldown, W1b) whose primary-source citations appear in §8.15 and the `graphcompat` module (emergent-gravity DM invisible collective; caps hand-assigned).

Derived ($w = \{-1, 0, 1/3\}$ via CC3/CC4/CC4’, `native_decide` on `fracton_sm_singlet`)

MCC (`torsion_channels_distinct_sources_distinct`; channel ma)

MCC (`taxonomy_channel_distinctness`)

MCC (`empirical_hook_ranking_strict`; priorities hand-assigned)

MCC (`phase5x_candidates_viability_matrix`; flags hand-assigned)

Derived (fig-cdm obstruction from traceless $T^{\mu\nu}$)

MCC (fracton bullet-cluster signature is strengthened results)

MCC (fracton cosmological signature is strengthened results)

Empirical (candidate building. Paper 17 exports those theorems into the DM/DE literature, with four measurable consequences.)

Empirical (literature)

Empirical (W1b)

Heuristic (W3 memo; $\mathcal{O}(1)$ prefactor Phase 6)

First, the hidden-sector classification is a falsifiable constraint surface; any future discovery of a non-singlet candidate violating $\mathbb{Z}_{16}$ charge +3 would invalidate the anomaly-driven motivation. No such discovery is indicated by LZ, XENONnT, or DARWIN projections—every candidate in our program is expected, on the strength of deep-research per-candidate bounds collected

in Table II (MCC entries), to be invisible to nuclear-recoil direct detection. Strengthening these bounds from classification-level (MCC) to derivation-level is a Phase 6 target.

Second, the ADW Λ -magnitude result reproduces the meV scale of $\rho_{\text{vac}}^{1/4}$ without tuning. We withdraw the prior “20%-agreement” quantification pending a Phase 6 full- V_{eff} calculation of the $\mathcal{O}(1)$ coefficient; for now the claim is scale-matching (“the meV scale emerges naturally from the condensation equilibrium, without fine-tuning”). We honestly separate this scale-matching from the withdrawn KV dynamics claim and note the QCD-topological-DE pathway as a DESI-compatible alternative.

Third, the SFDM cluster-merger forecast is the Paper 17 money plot and fills a confirmed gap in the most recent comprehensive BK review. It is the only Phase 5x observable with a specific detection timeline: first 3σ by ~ 2028 .

Fourth, fracton dark matter—viable conditional on the p-wave phase and a Dark QCD UV completion with $\mu \gtrsim \text{MeV}$ at BBN—is structurally compatible with every other Phase 5x sector (CC1 through CC7) and can explain core-cusp and diversity simultaneously through the $z=4$ subdiffusion mechanism.

What remains open: the MC extension to $L = 10, 12, 16$ is the gate for the vestigial-relic program, the

Phase 6 bordism infrastructure is the gate for discharging the Z16-vestigial charge hypothesis, and the Phase 6 hard theorems L4/L5 (MOND + FDR) are the gate for the dissipative-corrections section of SFDM.

The unifying theme of this paper is *structural*: every emergent-gravity DM candidate in our program is SM-gauge singlet and invisible to nuclear-recoil direct detection. This is itself a thematic prediction—if dark matter originates in an emergent-gravity sector, null direct-detection results at DARWIN are the expected outcome, and observable signatures must be sought in gravitational observables (cluster mergers, weak lensing, galaxy kinematics) rather than laboratory recoil experiments.

ACKNOWLEDGMENTS

We thank the Lean 4 and Mathlib communities for infrastructure and the authors of the BK 2025 review for focused attention on phenomenological gaps. Special credit to the Wan–Wang, Wang Ultra-Unification, Volovik Emergent-Gravity, Berezhiani–Khoury Superfluid-DM, Fang–Gu Topological-Gravity, Pretko–Nandkishore Fracton, and Kapustin–Spodyneiko / Jensen–Raz / Głódkowski p-wave programs whose results Paper 17 synthesizes.

-
- [1] W. G. Unruh, Phys. Rev. Lett. **46**, 1351 (1981).
 - [2] M. Visser, Class. Quant. Grav. **15**, 1767 (1998), arXiv:gr-qc/9712010.
 - [3] M. Crossley, P. Glorioso, H. Liu, JHEP **09**, 095 (2017).
 - [4] P. Glorioso, M. Crossley, H. Liu, JHEP **09**, 096 (2017).
 - [5] G. E. Volovik, JETP Lett. **82**, 319 (2005).
 - [6] F. R. Klinkhamer and G. E. Volovik, Phys. Rev. D **77**, 085015 (2008).
 - [7] L. de Moura and S. Ullrich, CADE **28**, 625 (2021).
 - [8] The mathlib Community, CPP **2020**, 367 (2020).
 - [9] Z. Wan, J. Wang, and Y. Zheng, Phys. Rev. D **100**, 085012 (2019).
 - [10] J. Wang, Phys. Rev. Research **2**, 013189 (2020).
 - [11] J. Wang, Mod. Phys. Lett. A **36**, 2130008 (2021).
 - [12] I. García-Etxebarria and M. Montero, JHEP **08**, 003 (2019).
 - [13] A. A. Abrikosov Jr. and F. R. Klinkhamer, Phys. Rev. D **99**, 105009 (2019).
 - [14] D. Diakonov, Mod. Phys. Lett. A **26**, 2439 (2011).
 - [15] L. Berezhiani and J. Khoury, Phys. Rev. D **92**, 103510 (2015), arXiv:1507.01019.
 - [16] L. Berezhiani, G. Cintia, V. De Luca, and J. Khoury, Phys. Rep. (2025) arXiv:2505.23900.
 - [17] L. Berezhiani, J. Khoury, and J. Wang, Phys. Rev. D **95**, 123530 (2017).
 - [18] DESI Collaboration, arXiv:2404.03002.
 - [19] DESI Collaboration, arXiv:2503.14738.
 - [20] J. Solà Peracaula, Class. Quant. Grav. **40**, 215004 (2023).
 - [21] L. Van Waerbeke and A. R. Zhitnitsky, “QCD topological dark energy,” arXiv:2506.14182 (2025).
 - [22] Planck Collaboration, A&A **641**, A6 (2020).
 - [23] X.-Y. Fang and Z.-C. Gu, arXiv:2106.10242.
 - [24] M. Pretko, Phys. Rev. B **95**, 115139 (2017).
 - [25] R. M. Nandkishore and M. Hermele, Annu. Rev. Condens. Matter Phys. **10**, 295 (2019).
 - [26] J. Shen *et al.*, Phys. Rev. B **106**, 045103 (2022).
 - [27] V. Krishna *et al.*, Phys. Rev. B **109**, 125131 (2024).
 - [28] A. Kapustin and L. Spodyneiko, “Hohenberg-Mermin-Wagner-type theorems and dipole symmetry,” Phys. Rev. B **106**, 245125 (2022), arXiv:2208.09056.
 - [29] K. Jensen and A. Raz, Phys. Rev. Lett. **132**, 071603 (2024).
 - [30] A. Głódkowski, F. Peña-Benítez, and P. Surówka, “Dissipative fracton superfluids,” arXiv:2401.01877 (2024).
 - [31] B. Feistl, K. Schraven, and S. Warzel, arXiv:2601.23078 (2026).
 - [32] SK-EFT Hawking Program, Paper 8, “FractonNon-Abelian formalization: Yang–Mills incompatibility of fracton topological phases” (2025).